



Hassan Bahar	F2023266769
Hassan Kareem	F2023266768
Subject:	Data Mining
Section:	V2
Topic:	House Price Prediction
Due Date:	27/Jan/2026
Submitted to:	Mrs.Rafia Asad

INTRODUCTION	1
Problem Statement and Objectives	2
Dataset Description	3
Data Preprocessing	4
Methodology	5
Model Selection	6
Results and Discussion	7
Business Implications	8
Conclusion	9
Future Work	10

House Price Prediction Using Data Mining Techniques

1. Introduction

The rapid growth of the real estate industry has increased the need for accurate house price estimation. House price prediction helps buyers, sellers, and real estate companies make informed decisions. This project focuses on predicting house prices using data mining and machine learning techniques. A real-world housing dataset was used, and different regression models were applied to achieve accurate predictions.

2. Problem Statement and Objectives

Problem Statement:

To predict the selling price of a house based on its features such as area, number of bedrooms, bathrooms, property type, city, and location.

Objectives:

- To analyze housing data and understand important factors affecting house prices
- To apply data preprocessing techniques to clean and prepare the data
- To train and evaluate multiple machine learning models
- To select the best-performing model for house price prediction

3. Dataset Description

A publicly available housing dataset was used for this project. The dataset contains real estate listings including information about property size, number of rooms, location, and price. Only sale properties were selected for analysis to maintain consistency.

Main features in the dataset include:

- Area_in_Marla
- Bedrooms
- Bathrooms
- Property Type
- City
- Location
- Price (Target Variable)

Dataset Preview:

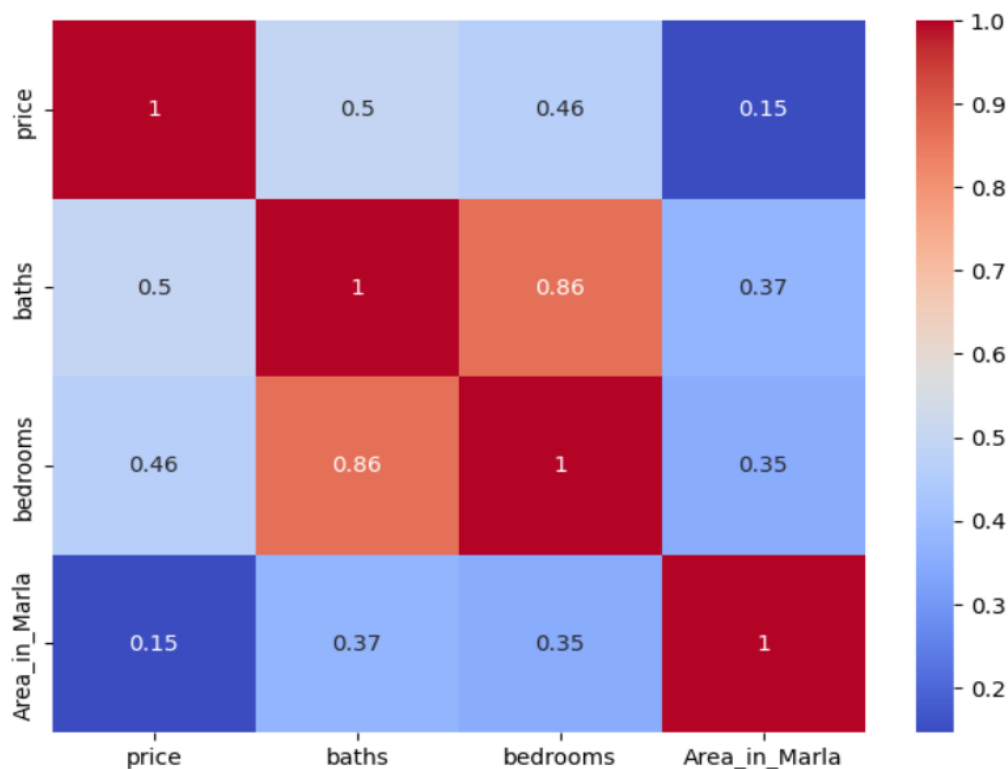
...	Unnamed: 0	property_type	price	location	city	baths	purpose	bedrooms	Area_in_Marla
0	0	Flat	10000000	G-10	Islamabad	2	For Sale	2	4.0
1	1	Flat	6900000	E-11	Islamabad	3	For Sale	3	5.6
2	2	House	16500000	G-15	Islamabad	6	For Sale	5	8.0
3	3	House	43500000	Bani Gala	Islamabad	4	For Sale	4	40.0
4	4	House	7000000	DHA Defence	Islamabad	3	For Sale	3	8.0

4. Data Preprocessing

Several preprocessing steps were applied to prepare the data for modeling:

- Removal of irrelevant columns (e.g., index column)
- Filtering the dataset to include only sale properties
- Handling missing values using appropriate techniques
- Exploratory Data Analysis (EDA) using descriptive statistics and correlation analysis
- One-Hot Encoding applied to categorical variables such as city, location, and property type
- Feature scaling using StandardScaler

Heat Map of Correlation:



5. Methodology

The project followed a standard data mining methodology:

1. Data Collection and Understanding
2. Data Preprocessing and Feature Engineering
3. Model Training using Linear Regression and Random Forest Regression(Default, Tuned)
4. Model Evaluation using MAE, MSE, and R^2 Score
5. Model Comparison and Final Model Selection

Random Forest Regression was selected as the final model due to its ability to capture non-linear relationships.

6. Models Used and Justification

Apriori Algorithm for Pattern Discovery

In addition to regression-based models used for house price prediction, the Apriori algorithm was applied as an additional data mining technique to discover hidden patterns and associations among property features. Since Apriori works on categorical data, continuous variables such as price and area were converted into categorical ranges. The extracted association rules provided valuable insights into how combinations of features such as location, property type, and price category are related.

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
14	(Bahria Town Karachi)	(Karachi)	0.087544	0.383526	0.087544	1.0	2.607387	1.0	0.053969	inf	0.675621	0.226262	1.0	0.614131
48	(3, Bahria Town Karachi)	(Karachi)	0.061454	0.383526	0.061454	1.0	2.607387	1.0	0.037885	inf	0.656840	0.160235	1.0	0.580118
135	(3, Medium, Bahria Town Karachi)	(Karachi)	0.057635	0.383526	0.057635	1.0	2.607387	1.0	0.035530	inf	0.654178	0.150276	1.0	0.575138
130	(House, 3, Bahria Town Karachi)	(Karachi)	0.050347	0.383526	0.050347	1.0	2.607387	1.0	0.031038	inf	0.649158	0.131275	1.0	0.565638
82	(House, Bahria Town Karachi)	(Karachi)	0.055436	0.383526	0.055436	1.0	2.607387	1.0	0.034175	inf	0.652655	0.144542	1.0	0.572271

The results of the Apriori algorithm revealed strong associations between certain property characteristics. For example, properties located in specific areas showed a strong relationship with higher price categories. These patterns help in understanding market behavior and complement the predictive models by providing interpretability and business insights.

Linear Regression was used as a baseline model due to its simplicity and interpretability. Random Forest Regression was applied to improve accuracy as it handles complex patterns and reduces overfitting. Then I changed the size of trees and changed some other parameters of this and retrained the model, but it did not show the big performance.

Random Forest parameters were selected based on performance comparison and experimentation.

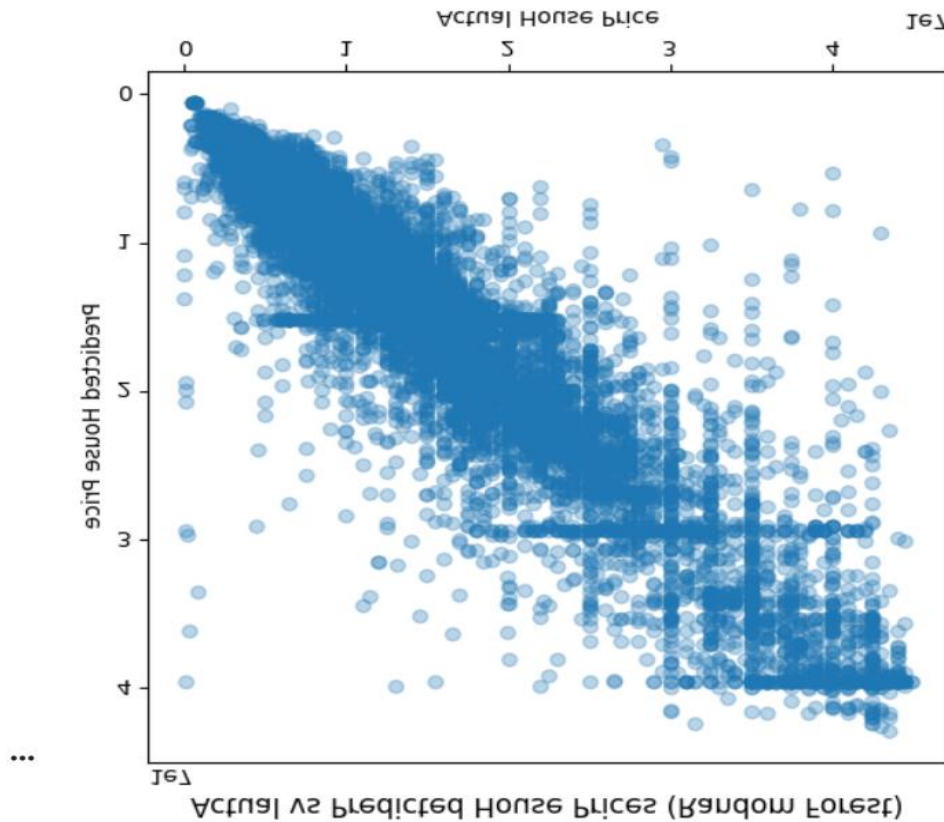
Comparison:

Model	MAE (↓ Better)	MSE (↓ Better)	RMSE (↓ Better)	R ² Score (↑ Better)
Linear Regression	3,598,197	1.71×10^{13}	4,136,000	0.703
Random Forest (Default)	2,270,143	1.33×10^{13}	3,650,000	0.865
Random Forest (Tuned)	2,544,340	1.40×10^{13}	3,742,000	0.844

7. Results and Discussion

The Linear Regression model achieved an R² score of approximately 0.70, while Random Forest Regression [Default] significantly improved performance with an R² score of approximately 0.86 and lower MAE. While Random Forest Regression [Tuned] showed R square score of 0.84.

These results indicate that Random Forest is more suitable for house price prediction due to its ability to model complex feature interact



ions.

8. Final Model Selection

Based on evaluation metrics, Random Forest Regression was selected as the final model. The trained model was saved and later used to predict house prices for new user inputs.

9. Business Implications

The developed model can be used by real estate companies to estimate property prices, help buyers avoid overpricing, and assist sellers in setting competitive prices. It can also support decision-making in real estate investments.

10. Conclusion

This project successfully demonstrated the use of data mining techniques for house price prediction. Through proper preprocessing and model selection, accurate predictions were achieved using Random Forest Regression.

11. Future Work

Future improvements may include:

- Incorporating additional features such as nearby facilities
- Hyperparameter tuning for further accuracy improvement
- Applying advanced models like Gradient Boosting or XGBoost
- Deploying the model as a web-based application